Gradient-descent trajectories on a loss surface

update $\boldsymbol{x}^{(t+1)} = \boldsymbol{x}^{(t)} - \eta \nabla f(\boldsymbol{x}^{(t)})$ on the bowl $f(\boldsymbol{x}) = \frac{1}{2}(a\,x_1^2 + b\,x_2^2), a \ll b$